Department of Electrical and Computer Engineering
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Project Title:
Understanding the Gaming Behavior of Adults

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ABSTRACT

This study explores the gaming behaviors of adults, a group of people aged 18 or above. This study aims to identify an adult's gaming behavior as concerning or non-concerning and through this, evaluate its association with a range of symptoms related to psychological, social, and physical health. The study also enlightens the motives, preferences, and effects of gaming on persons aged 18 or above. It does this by combining quantitative data from surveys with qualitative insights from many other research papers. Key findings show that individuals mostly engage in gaming for relaxation, social interaction, sometimes to escape from people or situations, and cognitive stimulation, with preferences varying significantly based on age, gender, and lifestyle. The study looks at the possible advantages and disadvantages of gaming for this age range, including how it affects social interactions, mental health, and time management. This research can help mental health practitioners, lawmakers, and game creators build environments that are supportive of adult gamers by shedding light on the complex gaming behaviors of adults. This research would help to set a margin for individuals so that their gaming behavior stays non-concerning. To improve adult gaming and maintain gaming as a positive and constructive activity, the paper ends with suggestions for future research and useful applications.

1. INTRODUCTION

In recent years, the landscape of digital entertainment has been significantly shaped by the rise of gaming, with video games becoming a prevalent and influential form of leisure activity among adults. Despite the common perception of gaming as a teen-only pastime, a significant portion of the adult population currently engages in various forms of gaming. Online gaming statistics will help to understand the growth of the gaming industry and the increasing engagement of adults in this expanding sector. According to a comprehensive list of the most relevant research-backed online gaming statistics, there are 3.2 billion video gamers in the world and 1.17 billion of them play online [1]. Online gaming revenue grew from $135.4 billion in 2020 to $176.06 billion in 2023. The global gaming industry was worth $214.20 billion in 2021, and $181.72 billion is attributed to online gaming. The global gaming industry is expected to be worth $504.9 billion by 2030. Asia is the biggest online gaming market with 1.48 billion gamers. 76% of gamers are over the age of 18 [1]. So, it may be assumed to be for teenagers but nowadays we can see a large number of adults are striking the gaming market. This trend warrants a comprehensive examination of the factors driving gaming behavior in adults, the impact of gaming on their lives, and the broader implications for society and the gaming industry.

Understanding the gaming behavior of adults is very important because it can affect many fields. It has a noticeable impact on social and psychological well-being. It can help in stress relief and improve mental health and is also a good way of doing social interaction. But surprisingly it can also lead to addiction and negative health outcomes. Nowadays, adults are also being driven toward gaming because they easily escape from social interaction along with negative emotions. So, understanding adult gaming behavior can help in creating policies and programs that promote healthy gaming habits and address addiction issues for awareness. Understanding their behaviors is crucial because adults represent a diverse demographic group of people with varying motivations, preferences, and gaming habits, influenced by factors such as career responsibilities, family dynamics, and social interactions.

This study aims to understand the gaming behavior of an adult and identify it as concerning or non-concerning. Understanding the gaming behavior of adolescents is a very popular topic for researchers but very few works are done on adult gaming behavior analysis. Mostly, research has shown a relationship between gaming addiction and mental health based on survey data where target outcomes are very specific like depression, addiction, etc., which fully relied on self-reported measures through a survey, which may introduce response bias and social desirability effects; potentially impacting the accuracy of the reported data. So, matching those records with IGD (Internet gaming disorder) symptoms is a bit risky. That’s why in our study, we simply calculated the weekly playing hours depending on individuals' daily lifestyle and labeled their gaming behavior as concerning or non-concerning so that it can generate basic awareness. According to a study by OXFORD, playing video games for 15-20 hours per week is starting to overplay, and playing more than 21 hours per week (or 3 hours per day) can begin to negatively affect the well-being of a person [2]. Though our study mainly focuses on the adult audience, this work will predict any individual's gaming behavior depending on their daily routine.

To assess and predict gaming behavior, this study included a set of questionnaires. Following the creation of the questionnaire, data was collected via Google Forms. From the collected dataset, data analysis and evaluation were done. A consent form was provided on the first page, and individuals were required to sign it to participate in this work. Major contributions of this work are illustrated below:

- A dataset has been collected using a Google form.
- The dataset has been preprocessed applying many techniques.

- 6 Machine learning algorithms have been applied to classify concerning or non-concerning gaming behavior.
- Hyperparameter optimization techniques are used to improve the performance of these models.
- An explainable AI tool, LIME, has been used to graphically explain the final predictions provided by machine learning models.

According to our knowledge after studying related works, this is the first time a newly developed dataset, of Bangladeshi participants, has been used to investigate adult gaming behavior.

The following is the order in which this article is organized. Section 2 shows the related works. Section 3 focuses on the methodology adopted in this work. Section 4 states the acquired outcomes. Section 5 discusses how the outcomes achieved our research objectives. Section 6 discusses the limitation of this work and finally, Section 7 includes a conclusion with remarks on our future work.

2. LITERATURE REVIEW

Gaming addiction is a very common condition nowadays characterized by excessive and compulsive use of video games that interferes with daily life. This addiction can have significant negative impacts on a person's mental, emotional, and physical well-being. Because of this serious negative impact, lots of research has been done in this field. This section briefly discusses some of the notable works in this area.

Related work on gaming addiction using machine learning

There is research done with a sample of 405 adolescents living in Kirikkale in Turkey where 231 are girls (57%) and 174 are boys (43%) [16]. Self-reported data were collected by questionnaire method. They evaluated Digital Game addiction with the digital game addiction (DGA) scale. Data analyses were performed using Python 3.9 software and SPSS 28.0 (IBM Corp., Armonk, NY, United States) package program. A logistic regression model was created based on the advanced feature selection method, and accuracy, sensitivity, specificity, FI-score, positive and negative predictive values were calculated with the help of the confusion matrix to evaluate the prediction performance of the model. They have obtained an accuracy of around 0.753 for their DGA (digital game addiction) prediction model.

There is another study aims to identify and understand the significant factors, such as perceived usefulness, attitude, and symmetric flow that influence the behavior intention in the online mobile game industry [14]. They collected symmetric data based on the questionnaire from a specified locality of India. Machine-learning techniques, such as logistic regression and SVM, are applied for further analysis and predictive insights. The evaluation is done among three parameters: perceived usefulness, attitude, and flow-over behavior intention. The logistic regression prediction outperforms, i.e., 88.67%, the other, which is 81.33% for SVM and 61.45% for ridge regression, in terms of perceived usefulness over BI. The study has several limitations: it only included individuals aged 16 to 35, excluding those above 35 where games are also popular.

Some of these recent works have been briefly presented below:

Table -1: Comparison of different research papers

Ref	Title	Sample size	Age range	Data collection mean	Advantage	Limitation
[3]	Problematic digital gaming behavior and its relation to the psychological, social and physical health of Finnish adolescents and young adults	293	13-24	Online survey	The research uses the Game Addiction Scale (GAS) to provide a thorough analysis of how gaming behavior affects different aspects of health and well-being.	This study was conducted with a sample of 293 respondents aged 13 to 24 years which generalizes the findings to a very small age range

[4]	Video games impact lifestyle behaviors in adults	292	Not explicitly stated	Online survey	The online format of the study enabled global participation, which enhanced the generalizability of the findings.	This study lacks a thorough examination of potential confounding variables that could influence the link between video game playing and lifestyle behaviors
[5]	Which health-related problems are associated with problematic video gaming or social media use in adolescents?	21,053	Students from secondary schools with a mean age of 14.4 years	Collected by two Municipal Health Services in the Netherlands in 2013-2014	The study included a large dataset from secondary schools, providing a robust dataset for analysis.	The study was conducted in the Netherlands with a specific age group, which may limit the generalizability of the findings to other populations or age ranges.

[6]	Internet Gaming Disorder: An Interplay of Cognitive Psychopathology	306	Not explicitly stated	Online survey	The study explores Internet Gaming Disorder (IGD) by analyzing how maladaptive gaming beliefs and psychological distress relate to IGD symptoms in young adults.	Relying on self-reported measures via a Google form survey can introduce response bias and social desirability effects, which might affect the accuracy of the data collected.

3. METHODOLOGY

This section explains the overall methods and techniques used to identify the gaming behavior of an adult as concerning or non-concerning. Figure - 1 shows the primary implementation steps taken in this study. Each step is explained below.

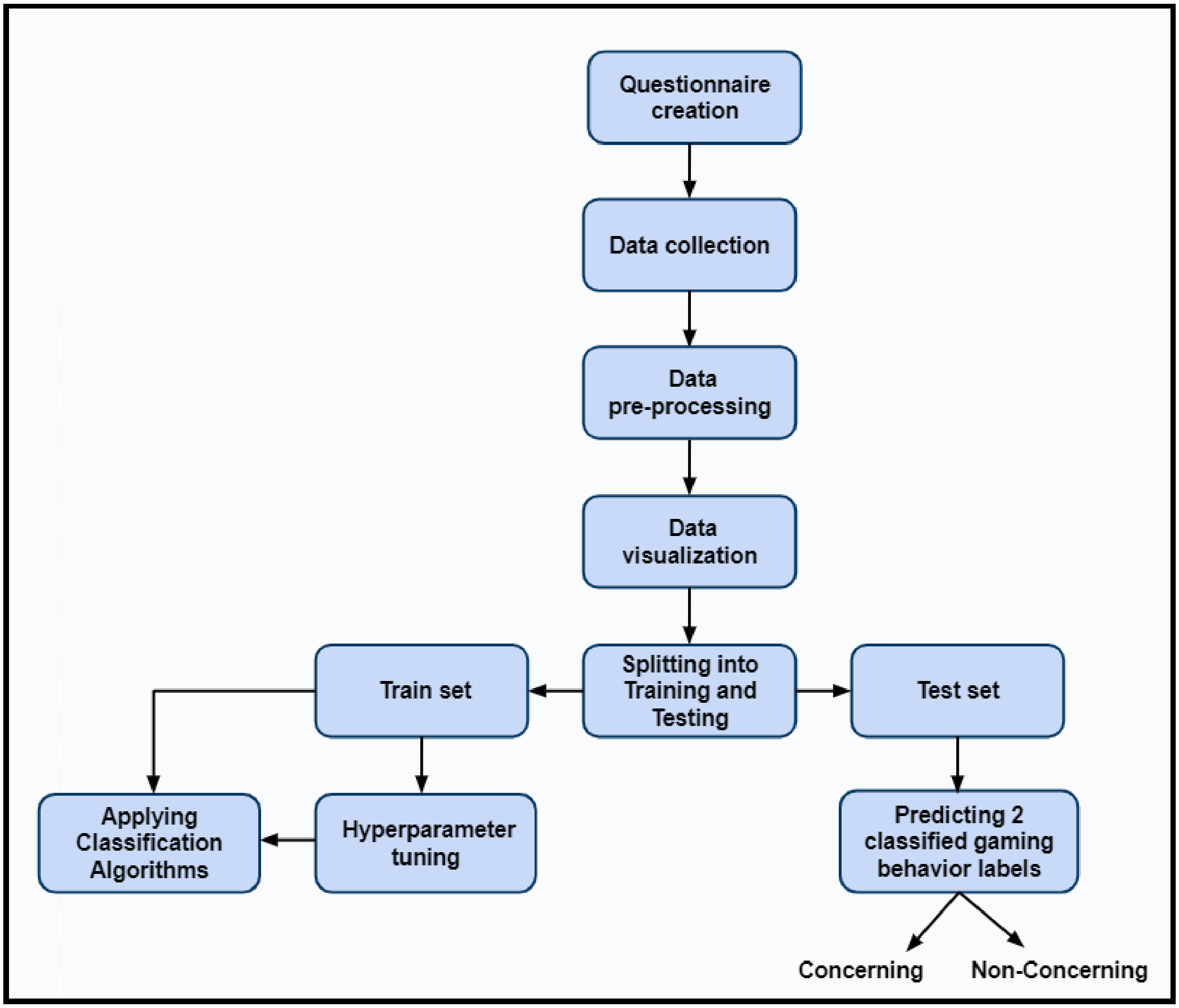


Figure - 1: Methodology of this study

3.1. Questionnaire creation

The creation of a dataset is one of the most important contributions of this study which is constructed from some basic questions, multiple questions related to an individual's daily gaming routine and gaming preferences, some relevant medical questions, some questions related to an individual's sleeping routine which are important to identify concerning gaming behavior of an individual. In total 30 questions were asked in this survey. The maximum competition time of the survey was 15 minutes.

3.2. Data collection

This study was conducted with a random sample of 141 respondents aged 18 or above. Participants completed an online survey using Google Forms. The survey was conducted over a timeframe of 5 weeks. The first page of the survey document contained a consent agreement to participate in the survey. A brief explanation of the study and their voluntary involvement in the consent form was provided to the participants. It was made clear that participants could stop taking the survey at any time and no traceable personal information would be collected. A total of 141 records were collected among those 13 records were submitted manually according to their given information while having a face-to-face conversation. The survey form was uploaded to some gaming groups from where the rest of the records were collected.

3.3 Data Pre-processing

The survey data has been preprocessed and analyzed for visualization, and machine learning models have been implemented on preprocessed data for identification as concerning and non-concerning gaming behavior. In the case of data preprocessing,

- We renamed the survey columns to avoid complexity in further model implementation. For example, we renamed the column “Division in Bangladesh you are based in:” to “Division”.
- Then we dropped the rows of participants who did not consent to participate in the survey and participants who did not play any video games.
- After shortlisting all necessary participants' data, we dropped unnecessary columns such as Timestamp, Comments, Most_played_games, etc. as they will not be needed for machine learning model implementation.
- Replaced null values of any column with the median of that column.
- Any column that contains multiple values is replaced by their average. For example: An individual who plays around 4-5 hours a day is replaced by 4.5 hours.
- For model implication it is flexible to work with numeric data rather than categorical. For that reason, we separated columns where needed so that we could get numeric data. For example: Which gaming mode do you play: online,

offline, or both? This output data is categorical. So, we made separate columns titled plays_online, plays_offline, and plays_both, where the outputs will be stored as 0/1 and after that dropped the previous column.

- We calculated Total_sleep_hours using two columns Sleep_time, Wakeup_time.
- We counted No_of_reasons_to_play, from collected Playing_reason column. counted Count_of_play_time from collected Play_time. counted No_of_Health_issues from collected Health_issues.
- For the columns that had only 'Yes' and 'No' values, we used Scikit Learns LabelEncoder.
- For the columns that had ordinal values, we used Scikit Learns OrdinalEncoder.
- We used one hot encoder for the attributes: Gender, Division, Education, Marital_status
- We used Scikit Learn Min-max scaler to do normalization on all the columns, so that the columns that have a bigger range of values do not affect the target label more and we can get all the values in the same range.

3.4 Data visualization

After data pre-processing, we showed how strongly each column is correlated with each other through a heatmap [Figure - 2]. This correlation between 2 variables indicates that there is a connection, relationship, or dependency between them; for which a change in one variable affects another. We have also used pandas profiling which generates an interactive HTML report that includes statistical summaries, data visualizations, and various metrics. We have shown a figure depicting the names of the most played games by the participants. We have also shown scatter plots of different columns of the dataset to better visualize the data.

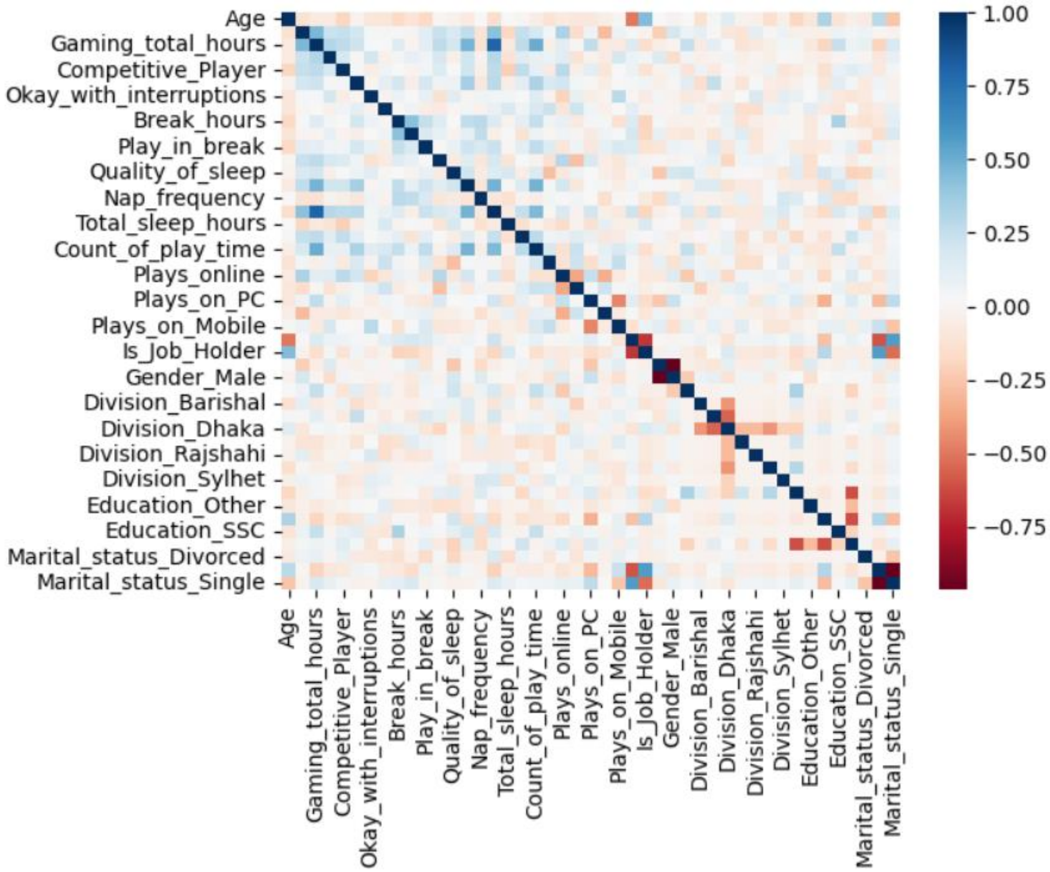


Figure - 2: Heatmap after data pre-processing

Figure – 3: The most played games by the participants

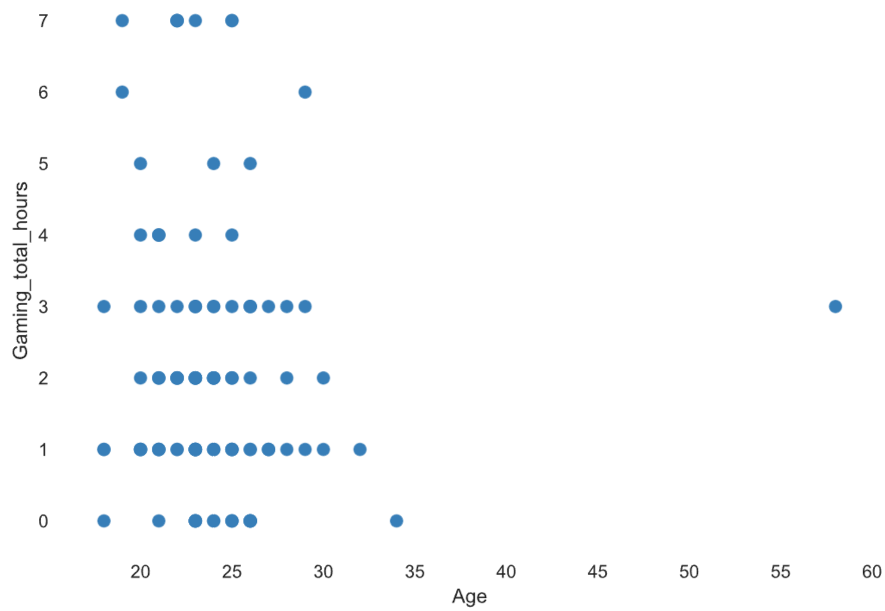


Figure – 4: Scatter plot for age and total gaming hours of the participants

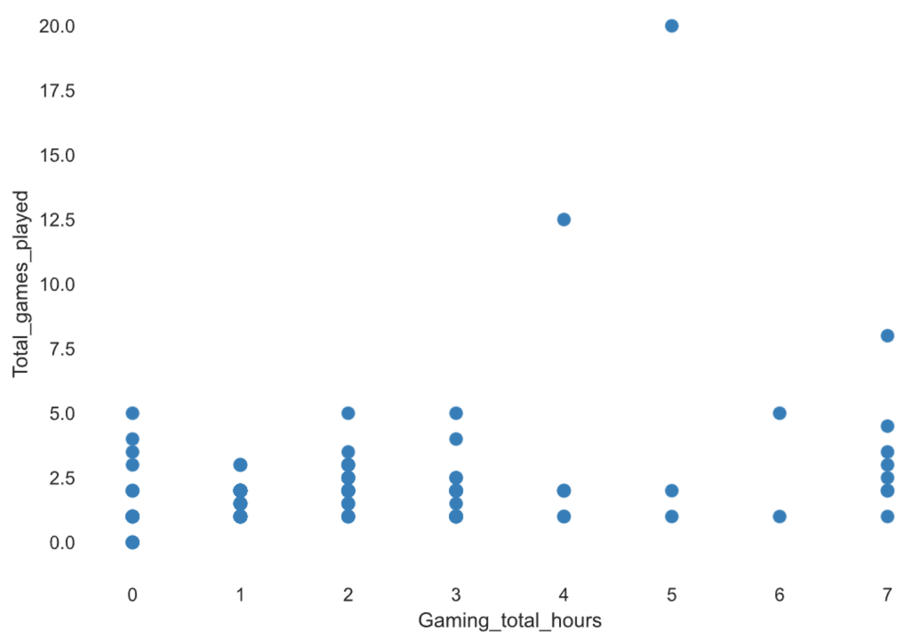


Figure – 5: Scatter plot for 'total gaming hours' and 'total games played by the participants'

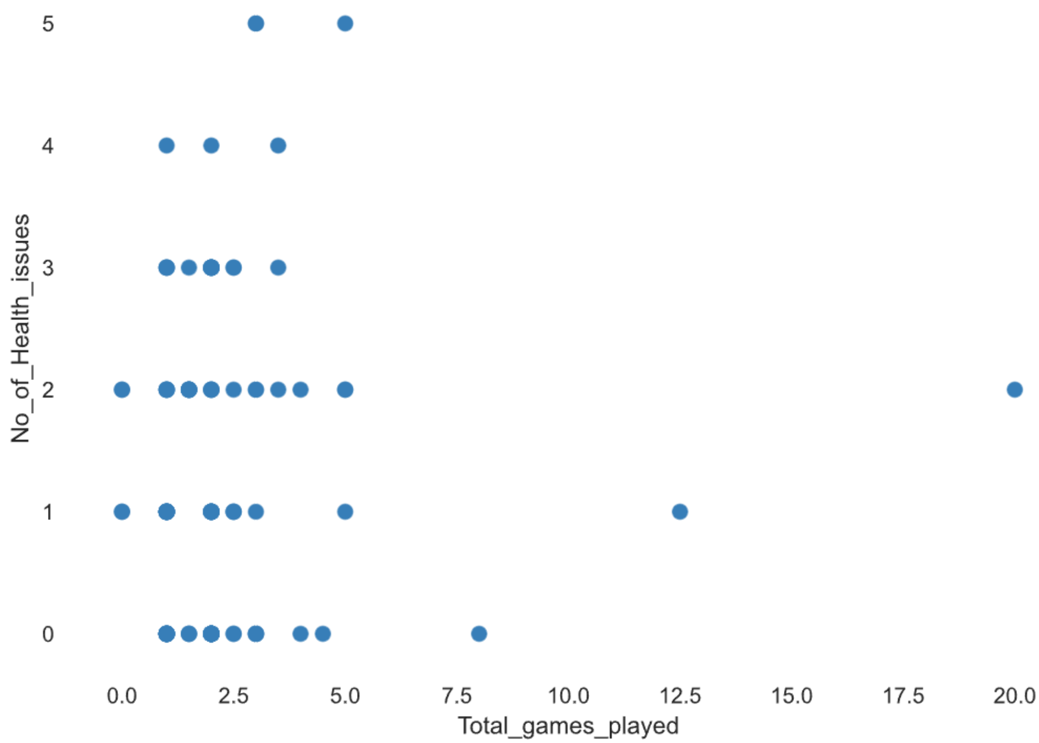


Figure – 6: Scatter plot for total games played and number of health issues of the participants

3.5 Data Division

After doing data pre-processing, we split the dataset into 2 sets.

1. Training set and
2. Test set.

From the entire dataset, we separated 20% data for testing, and the rest of the 80% we used for training the model. We've added the stratify parameter and set it to y (true label/target variable). This ensures that the distribution of the 2 classes will be the same as the original dataset in the train and test sets.

3.6 Hyperparameter Tuning

Here, we used GridSearchCV() which is a hyperparameter optimization technique, used to find the best values for the hyperparameters [criterion, splitter, max_depth, min_samples_split, min_samples_leaf, and max_features] for KNN, Decision Tree, Random Forest, and Ada Boost. In GridSearchCV() all possible combinations of hyperparameters are tested, so the best combination is guaranteed to be found.

3.7 Classification Models Application

After preprocessing the data, 115 records were obtained in the final dataset. The stratified approach was used to split the dataset into train set (80%), and test set (20%), so that the two datasets contain equal representations from both concerning and non-concerning gaming behavior. We've added random states so that the same result can be reproduced whenever the models are run. According to a study by OXFORD, playing video games for 15-20 hours per week is starting to overplay, and playing more than 21 hours per week (or 3 hours per day) can begin to negatively affect the well-being of a person [2]. We used this analysis to find 'y_true' which represents the actual value or true label. Here we created a function that will check if gaming frequency is 'Rarely' or 'Once a week' it will return 0 which means non-concerning gaming behavior. But if gaming frequency is 'Several times a week' then it will check if the total gaming hour on a day is between 4 to 7 hours or more, if yes then it will return 1 (which means concerning gaming behavior). Now if the gaming frequency is 'Daily' then it will check if the total gaming hours daily is between 3 to 7 hours or more, if yes then it will return 1 (which means concerning gaming behavior), or else 0 (which means non-concerning gaming behavior). Figure - 7 shows the y_true labeling steps.

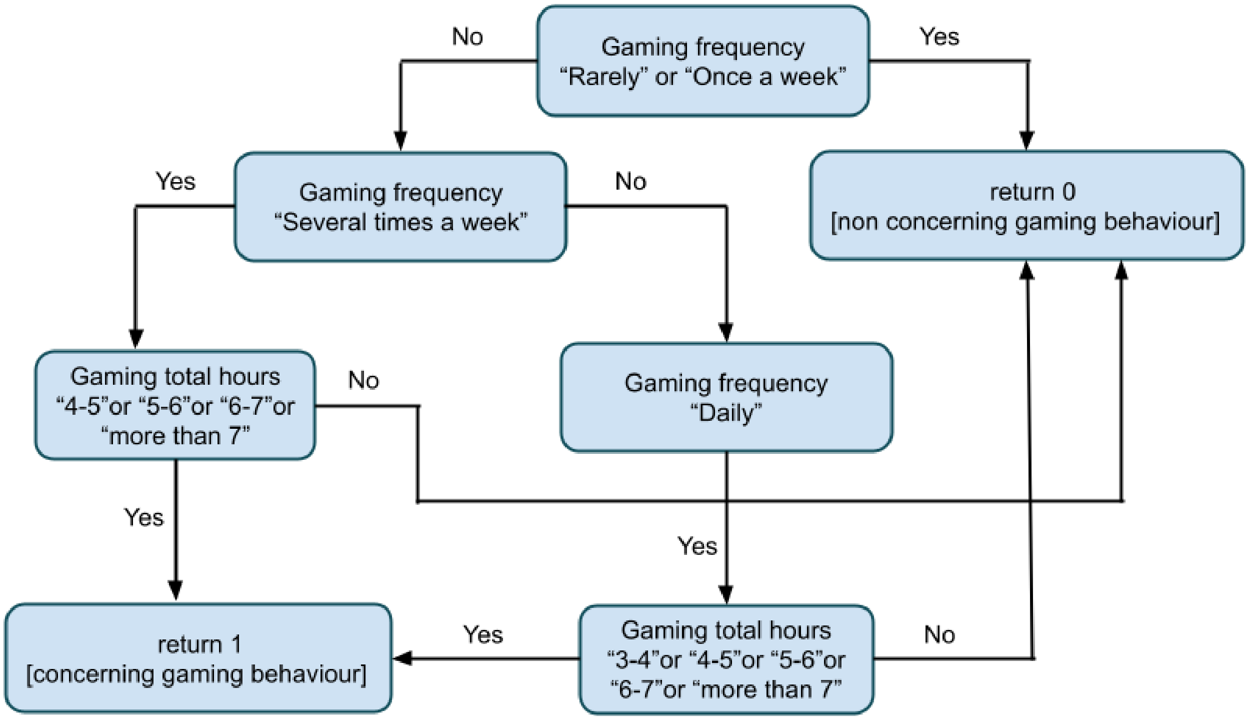


Figure - 7: Steps of labeling concerning and non-concerning gaming behavior.

6 machine learning models (KNN, Decision Tree, Logistic Regression, Support Vector Machine classifier, Random Forest, and Ada Boost) are applied in this study. A dummy classifier (Scikit Learn Dummy classifier with strategy as ‘most frequent’) is also applied to measure the performance of all other classifiers.

4. RESULTS

The performance metrics used to measure the models' performance were: Accuracy, Precision, Recall, and F1-Score. The formulas for these performance measures are given below:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

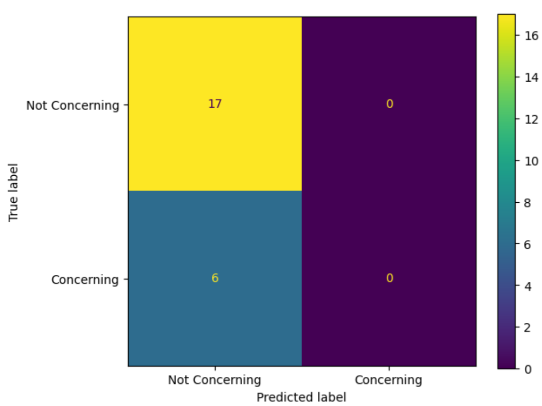
$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

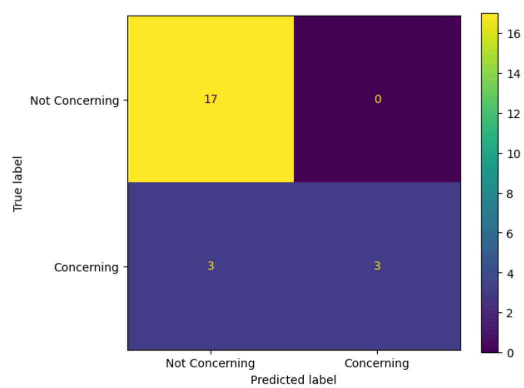
1. Confusion Matrix:

The Confusion Matrices of the models are given below:

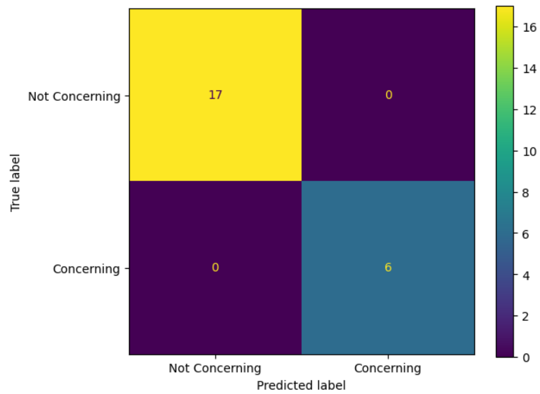
(i) Base Classifier



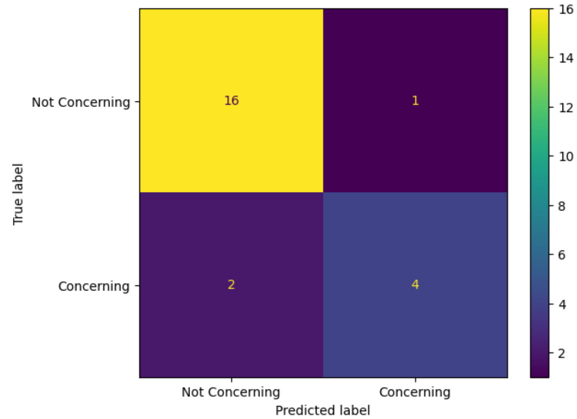
(ii) KNN with default parameters



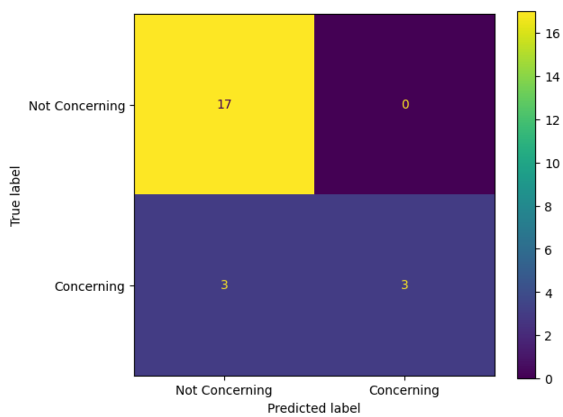
(iii) Decision Tree with default parameters



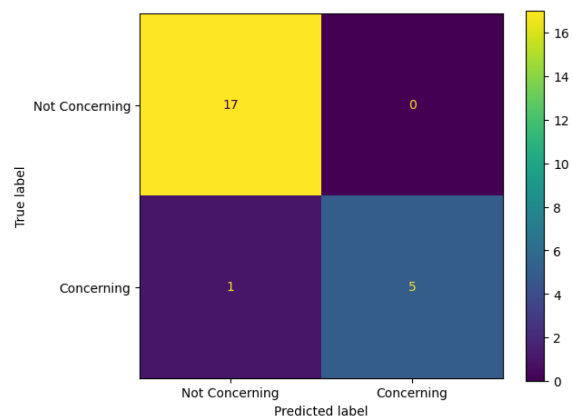
(iv) Logistic Regression with default parameters



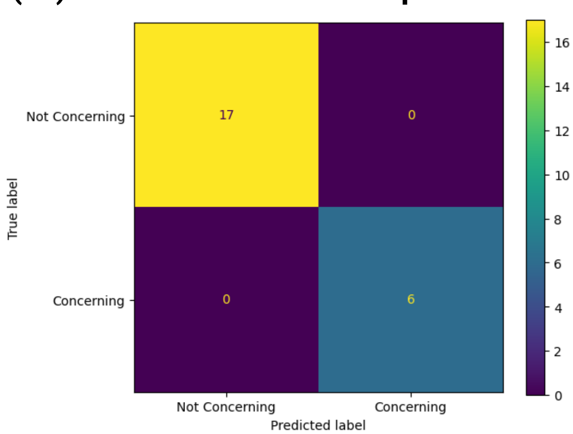
(v) SVC with default parameters



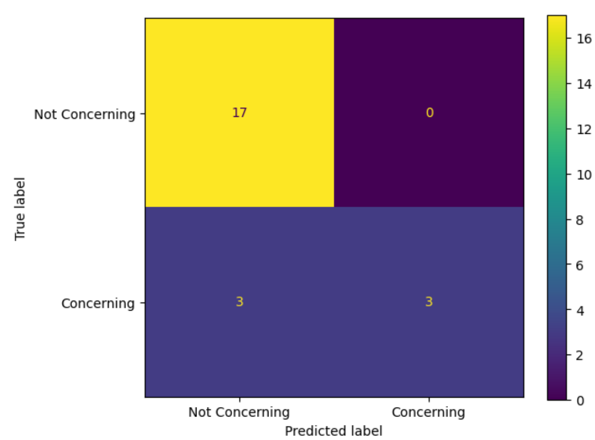
(vi) Random Forest with default parameters



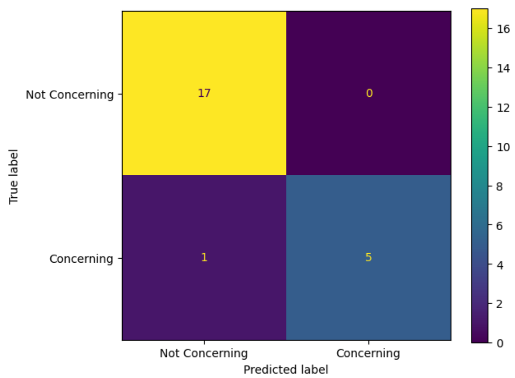
(vii) AdaBoost with default parameters



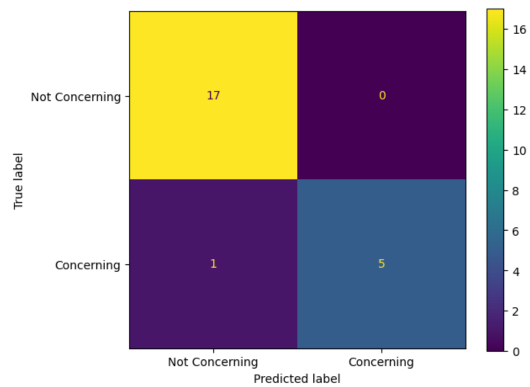
(viii) KNN with k=7 and weights=uniform



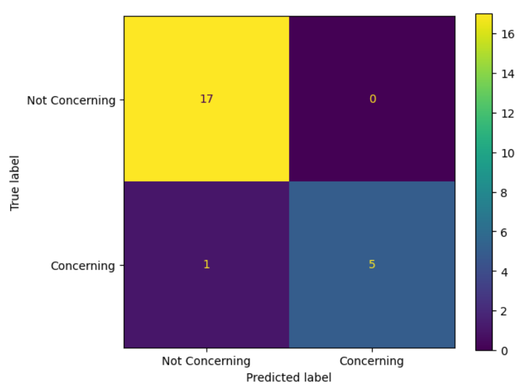
(ix) Decision Tree with criterion=entropy, max_depth = 6, max_features= log2, min_samples_leaf= 1, min_samples_split= 10, splitter=best



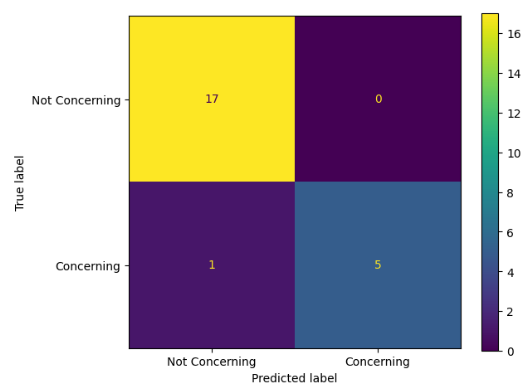
(x) Random Forest with criterion=entropy, max_features=log2, min_samples_leaf=2, n_estimators=17



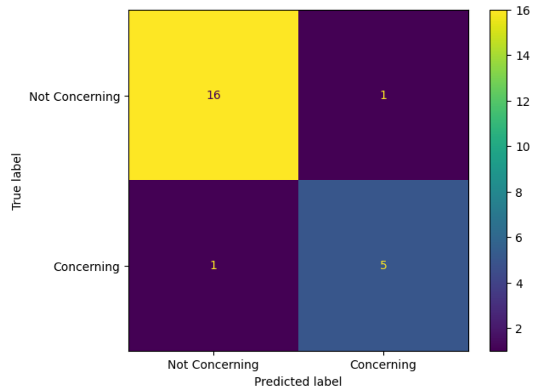
(xi) AdaBoost with algorithm=SAMME, estimator=DecisionTreeClassifier(random_state=100), learning_rate=1, n_estimators=1



(xii) AdaBoost with estimator=DecisionTreeClassifier(random_state=100), learning_rate=1, n_estimators=1



(xiii) AdaBoost with algorithm=SAMME.R, estimator=RandomForestClassifier(criterion=entropy, max_features=log2, min_samples_leaf=2, n_estimators=17), learning_rate=5, n_estimators=2



From the confusion matrices, (iii) and (vii) are the best models, as they have zero FN's.

2. **Train Accuracy and Test Accuracy:**

Table – 02: Train accuracy and Test accuracy of the ML models

Algorithms	**Train Accuracy**	**Test Accuracy**
i. Base Classifier	0.750000	0.739130
ii. KNN with default parameters	0.858696	0.869565
iii. Decision Tree with default parameters	1.000000	1.000000
iv. Logistic Regression with default parameters	0.934783	0.869565
v. SVC with default parameters	0.891304	0.869565
vi. Random Forest with default parameters	1.000000	0.956522

vii.	**AdaBoost with default parameters**	1.000000	1.000000
viii.	**KNN with k=7 and weights=uniform**	0.847826	0.869565
ix.	**Decision Tree with criterion=entropy, max_depth= 6, max_features=log2, min_samples_leaf= 1, min_samples_split= 10, splitter=best**	0.978261	0.956522
x.	**Random Forest with criterion=entropy, max_features=log2, min_samples_leaf=2, n_estimators=17**	1.000000	0.956522
xi.	**AdaBoost with algorithm=SAMME, estimator=DecisionTreeClassifier(random_state=100), learning_rate=1, n_estimators=1**	1.000000	0.956522
xii.	**AdaBoost with estimator=DecisionTreeClassifier(random_state=100), learning_rate=1, n_estimators=12**	1.000000	0.956522
xiii.	**AdaBoost with algorithm=SAMME.R, estimator=RandomForestClassifier(criterion=entropy, max_features=log2, min_samples_leaf=2, n_estimators=17), learning_rate=5, n_estimators=2**	1.000000	0.913043

Figure - 8: Bar Plot of Train Accuracy and Test accuracy of the ML models

Our experiment found that ML models (iii) and (vii) give a 100% Train and Test accuracy. Models (vi), (x), (xi), and (xii) give a 100% train accuracy and a 95.65% test accuracy.

3. Different Performance measures:

Table 03: Results of Different Performance Measures of the ML Models

Name of Algorithm	Accuracy	Precision	Recall	F1 Score	ROC AUC
i. Base Classifier	0.739130	0.000000	0.000000	0.000000	0.500000
ii. KNN with default parameters	0.869565	1.000000	0.500000	0.666667	0.750000
iii. Decision Tree with default parameters	1.000000	1.000000	1.000000	1.000000	1.000000
iv. Logistic Regression with default parameters	0.869565	0.800000	0.666667	0.727273	0.803922
v. SVC with default parameters	0.869565	1.000000	0.500000	0.666667	0.750000

vi.	**Random Forest with default parameters**	0.956522	1.000000	0.833333	0.909091	0.916667
vii.	**AdaBoost with default parameters**	1.000000	1.000000	1.000000	1.000000	1.000000
viii.	**KNN with k=7 and weights=uniform**	0.869565	1.000000	0.500000	0.666667	0.750000
ix.	**Decision Tree with criterion=entropy, max_depth = 6, max_features= log2, min_samples_leaf= 1, min_samples_split= 10, splitter=best**	0.956522	1.000000	0.833333	0.909091	0.916667
x.	**Random Forest with criterion=entropy, max_features=log2, min_samples_leaf=2, n_estimators=17**	0.956522	1.000000	0.833333	0.909091	0.916667
xi.	**AdaBoost with algorithm=SAMME, estimator=DecisionTreeClassifier(random_state=100), learning_rate=1, n_estimators=1**	0.956522	1.000000	0.833333	0.909091	0.916667
xii.	**AdaBoost with estimator=DecisionTreeClassifier(random_state=100), learning_rate=1, n_estimators=1**	0.956522	1.000000	0.833333	0.909091	0.916667
xiii.	**AdaBoost with algorithm=SAMME.R, estimator=RandomForestClassifier(criterion=entropy, max_features=log2, min_samples_leaf=2, n_estimators=17), learning_rate=5, n_estimators=2**	0.913043	0.833333	0.833333	0.833333	0.887255

We need higher Recall for a good Model to predict the gaming behavior, as gaming pattern concerning (TP) being predicted as non-concerning (FN) is a more grievous error. We need to lower FN’s, which in turn will make Recall higher. Our experiment found that models (iii), and (vii) give a 100% Recall.

4. **XAI using LIME (Local Interpretable Model-agnostic Explanations):**

(a) XAI for Default Decision Tree with criterion=gini, and splitter=best (with 10 features)

(For ML model iii)

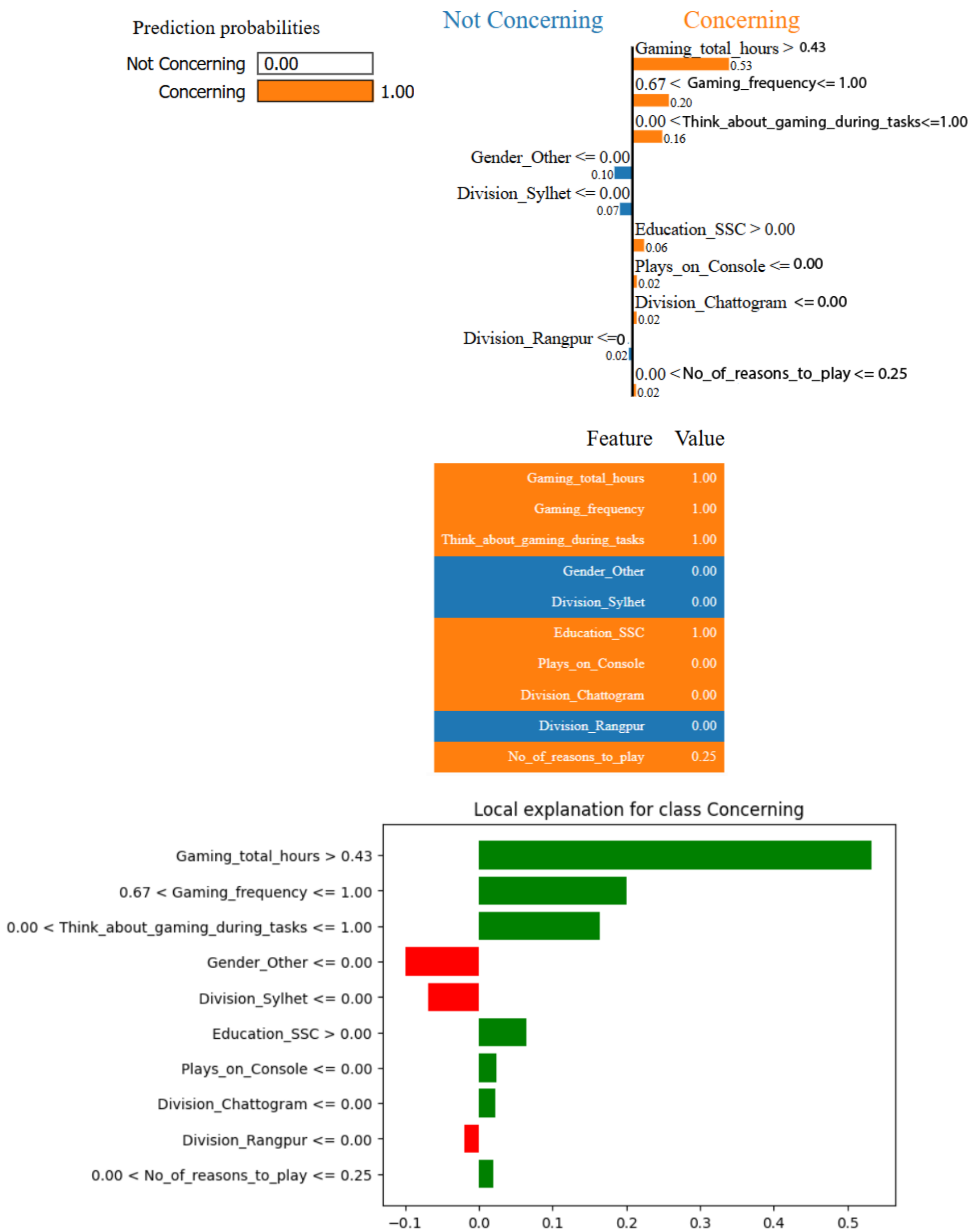


Figure - 9: Gaming pattern prediction for class ‘Concerning’ using LIME Explainable AI with ML model (iii)

(b) XAI for Default AdaBoost with n_estimators=50, learning_rate=1, algorithm=SAMME.R (with 10 features)

(For ML model vii)

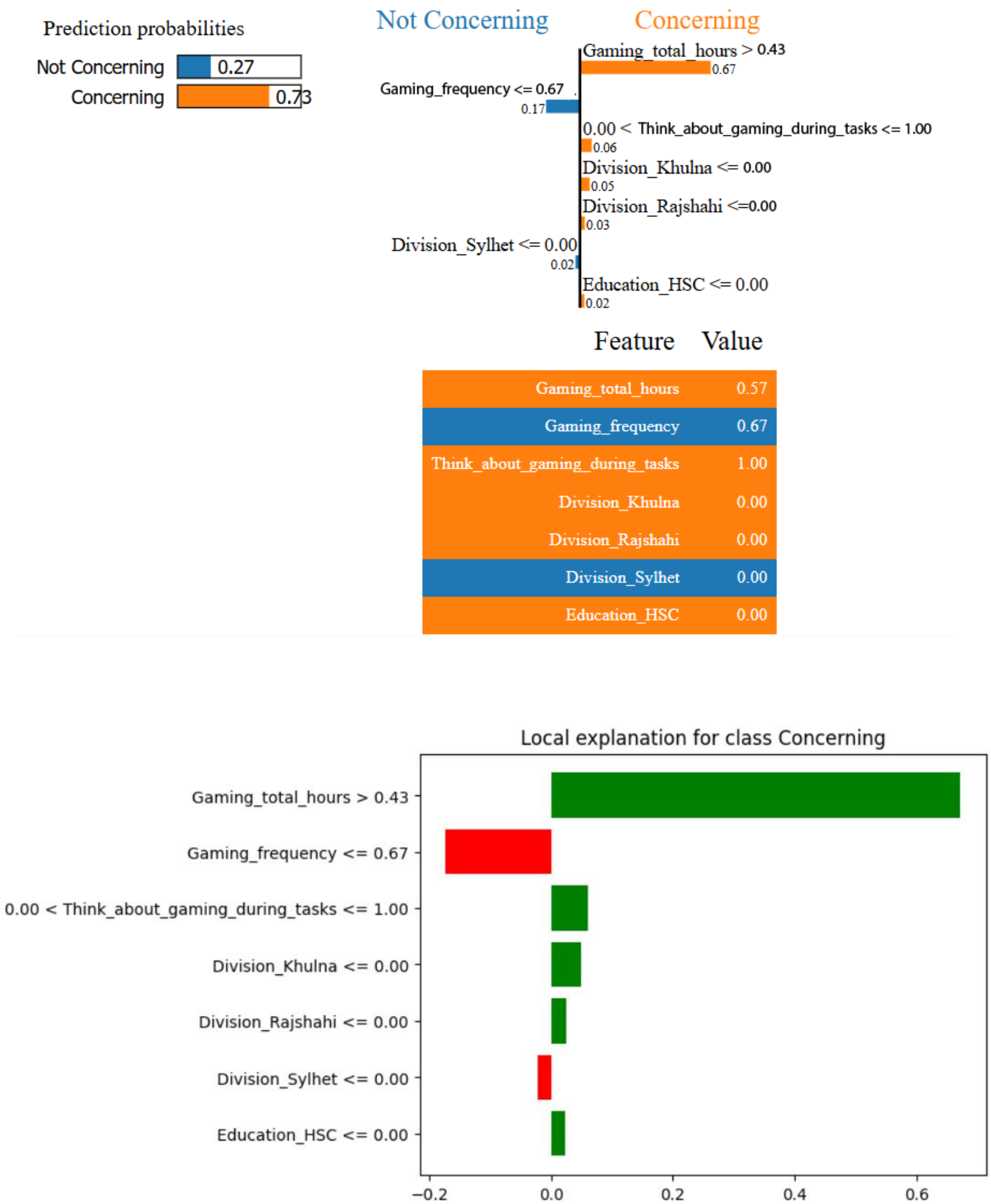


Figure -10: Gaming pattern prediction for class ‘Concerning’ using LIME Explainable AI with ML model (vii)

5. DISCUSSION

This study aims to identify whether the gaming behavior of adults is concerning or non-concerning. Internet Gaming Disorder (IGD) has been identified as a mental disorder by the American Psychiatric Association [7]. Gaming addiction is classified under IGD [8].

Our experiment found that the Decision Tree classifier and the AdaBoost Classifier (with default parameters) give perfect prediction models, with Accuracy, Precision, Recall, F1-score, and ROC AUC all at 100%. The reason for such high performance could be that we only had 115 data samples. A large sample size would have helped us to do better outcome analysis.

Table -04: Comparison of different machine-learning related works

Ref	Title	Dataset	Number of participants	Best model	Accuracy
[16]	Exploring obesity, physical activity, and digital game addiction levels among adolescents: A study on machine learning-based prediction of digital game addiction	Custom dataset of Turkish children [living in Kirikkale]	405	Logistic Regression Analysis	75.3%

[18]	A Machine Learning Based Approach to Predict Online Gaming Addiction in the Context of Bangladesh	Custom dataset of Bangladeshi youth	Not explicitly stated	AdaBoost	93%
	This work	Custom dataset of adults in Bangladesh	141	Decision Tree and AdaBoost	100%

6. LIMITATION

This study is conducted to understand adults' gaming behavior and predict whether it is concerning or not, using machine learning models. This study reached its destination along with some limitations.

- The number of collected data was very small (only 115 usable samples).
- After doing hyperparameter tuning, the accuracy and recall of the Decision tree and Ada Boost classifiers decreased in our study, the opposite of what is supposed to happen.
- We only used total gaming hours to find ‘y_true’, and ignored sleeping time and wakeup time, which can be potential parameters to identify a person's gaming pattern as concerning or non-concerning.
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7. CONCLUSION

This research paper worked with multiple machine learning algorithms implemented on a newly created dataset constructed by Bangladeshi participants for classifying the concerning or non-concerning gaming behavior of adults. This study will help to raise awareness among people, by understanding their gaming behavior, before it leads to any negative health issues or concerning daily routines. At present, we mainly focused on gaming frequency and total gaming hour parameters to classify gaming behavior. We have ignored other potential parameters that could lead to more accurate findings. Our future goal would be to encounter more potential parameters that hold valuable insights into identifying gaming behavior. We will continue to collect more data and apply more machine-learning models to find better results. We also plan to apply clustering algorithms on the collected data to identify any patterns that the data might have.

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